

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

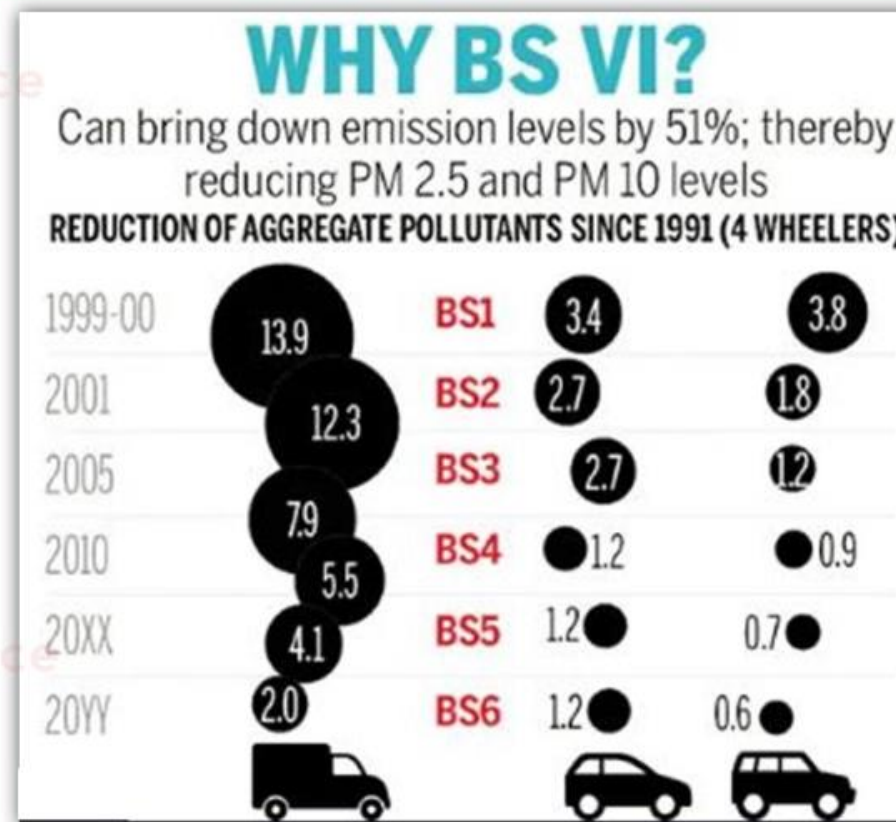
A COMPREHENSIVE COVERAGE

MODULE - 23

AIR POLLUTION

BS VI NORMS

- All vehicles sold after 1st April, 2020 will have to comply with BS VI standards. The new standards will require both automobile manufacturers and oil marketing companies to change their products.
- Rising number of vehicles and increasing air pollution have pushed the country to take a five-year leap in emission standards – Move directly from BS IV to BS VI.
- The introduction of BS VI vehicles will substantially narrow the gap between emission limits for petrol and diesel.
- Particulate emissions for diesel cars will be 82% to 93% lower than BS IV level, while Nitrogen Oxides (NOx) emissions will be 68% lower.



BS – VI NORMS

- For petrol engines emissions will be 25% lower. Particulate emissions for heavy duty vehicles will be 50 to 67% lower than the BS IV level.
- One of the Major differences between BS VI and BS IV is that BS VI contains 5 times less Sulphur content (10 PPM) as compared to BS IV (50 PPM).
- The government will also ensure that the automobile companies use the most efficient diesel particulate filter.
- BS VI compliant vehicles use sophisticated technologies like Diesel Particulate Filter and Selective Catalytic Reduction (SCR) Unit.
- The SCR unit reduces NO_x emissions by converting NO_x into nitrogen and water vapour. SCR uses a fluid which contains urea. When the fluid comes into contact with the exhaust gases, urea gets converted into CO₂ and Ammonia. The ammonia converts NO_x into water vapour and nitrogen.



EMISSION STANDARDS

DIESEL ENGINES (in g/Km)			PETROL ENGINES (in g/Km)		
BS IV		BS VI	BS IV		BS VI
0.50-0.74	Carbon Monoxide	0.50-0.74	1.0-2.27	Carbon Monoxide	1.0-2.27
0.30-0.46	HC+NO _x	0.17-0.215	0.1-0.16	HC+NO _x	0.1-0.16
0.25-0.39	Nitrous Oxides (NO _x)	0.08-0.125	0.08-0.11	Nitrous Oxides (NO _x)	0.060-0.082
0.025-0.06	Particulate Matter (PM)	0.0045	NA	Particulate Matter (PM)	0.0045

USAGE OF PEMS

- Vehicles that meet emission norms get certified for it in labs.
- But after the vehicles come on road, there is no check on the longevity of compliance to norms.
- To make technologies work till the vehicle's life on the road and to prevent frauds such as the Volkswagen scandal, Portable Emissions Measurement System (PEMS) will be used for vehicle certification from BS VI. This testing mechanism checks the Real-World Driving Emissions (RDE) from the vehicles.
- China, India, Mexico, Brazil, Russia and Australia have decided to implement Euro VI in the 2018-2020 timeframe.
- As India enforces BS VI standards, majority of the vehicles on the road will meet Euro VI norms by 2030 as new vehicles gradually replace the old ones.
- This will result in massive emissions reduction from light duty and heavy-duty vehicles. Once implemented, by 2030 Euro VI will reduce PM 2.5 emissions in the 6 countries by 74% of 2015 levels

EMISSIONS FROM THERMAL POWER PLANTS

- Emissions from coal fired thermal power plants can be categorized into three categories:
 - a) Gaseous Emissions – CO_2 , SO_2 , CO , NO_x and N_2O .
 - b) Particulate Matter – Sulphur particles and Fly Ash.
 - c) Volatile Organic Compounds and Trace Elements –
 - ✓ Mercury
 - ✓ Arsenic
 - ✓ Cadmium and
 - ✓ Lead (Toxic Heavy Metals).



ISSUES IN INDIA

- Thermal power plants are responsible for around 70% of total water use of all industries; they are responsible for over 60% of PM emissions, 50% of SO₂ and NO_x emissions and more than 80% mercury emissions by the industrial sector.
- In India, the EIA procedures practiced by the power plants are very old, which leads to underestimation of their impacts (especially due to long range transport of particulate matter).
- According to a report, the emission standards practiced in 2013 were at least 5-10 times worse than those practiced in countries like the US, European Union, China and Australia.



ISSUES IN INDIA

- Particulate matter from thermal power plants and industries around Delhi contributes to 35% PM 2.5 in winter in Delhi.
- Since Delhi has 13 thermal power plants with a capacity of over 11,000 MW within 300km radius, the emissions are often blown into NCR by the winds.
- Among the many steps taken to control air pollution in Delhi, curbing emissions from thermal power plants is strongly recommended.

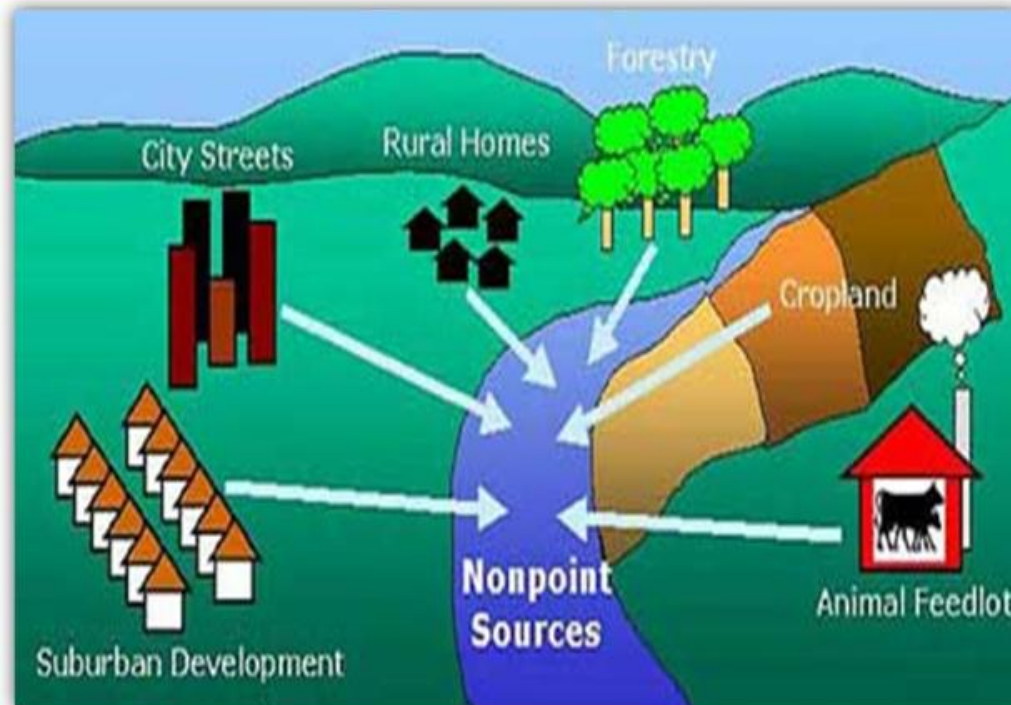


EMISSION STANDARDS FOR COAL-BASED POWER PLANTS

Thermal power plants are point sources of pollution, so we can actively control their emissions. This is why emission standards are important.

POINT SOURCE AND NON-POINT SOURCES

- Factories and power plants are a source of point source pollution.
- Non-point source pollution is the opposite of point source pollution with pollutants released in a wide area. Ex: A city during a thunderstorm. As rainwater flows over road, it washes away drops of oil leaked from car engines, particles of tire rubber, animal waste and trash. The runoff goes into a sewer and ends up in a nearby river.



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Runoff is a major cause of non-point source pollution.

EMISSION STANDARDS FOR COAL-BASED POWER PLANTS

POINT SOURCE AND NON-POINT SOURCES

- The MOEF notified emission norms in December 2015 for coal based thermal power plants.
- The standards were notified under the Environment (Protection) Amendment rules 2015.
- The standards cover particulate matter, SO_2 , NO_x , mercury and water use.
- The standards tightened the emissions for particulate matter. This long overdue revision in standards has happened after nearly 25 years.
- The implementation of 2015 standards will lead to reduction in PM, SO_2 , NO_x by 40 to 50%. It will also reduce mercury emissions by 60% and water consumption by another 40%.



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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588